NOTICE
OF REINSTATEMENT

MIL-G-29011(TD)
NOTICE 1
18 March 1985

MILITARY SPECIFICATION

GUIDES, OPERATION AND MAINTENANCE FOR TRAINING AIDS FOR PREPARATION OF

Military Specification MIL-G-29011(TD) dated 1 March 1971 and Amendment 1 dated 16 February 1977, is hereby reinstated and may be used.

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with the specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

Preparing activity:

Navy - TD

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Training Equipment Center, Code 424, Orlando, Florida 32813-7100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of the reinstated document or by letter.

FSC - TMSS

AMSC Number N3483

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MILITARY SPECIFICATION

GUIDES, OPERATION AND MAINTENANCE, FOR TRAINING AIDS
FOR PREPARATION OF

This amendment forms a part of Military Specification MIL-G-29011(TD), dated 1 March 1971.

PAGE 5

3.2.5.2 Delete and substitute the following:

"3.2.5.2 List of Maintenance Parts. All mechanical components that are subject to wear or deterioration and all electrical and electronic components shall be listed. Noun name, modifier, basic characteristics, part number, manufacturer's code, and reference designator or figure and index number of the part shall be included when applicable. Electrical and electronic components shall be listed first in alphanumeric order by reference designator, followed by all other parts listed in alphanumeric order by part number. The list shall be prepared as shown in figure 2 of this specification. For the purpose of this specification, an electrical and electronic component is defined as any device or piece part, such as a coil, resistor, capacitor, transistor, vacuum tube, or generator, having distinct electrical characteristics and having terminals at which it may be connected to other components to form a circuit."

Custodian:
Navy-TD

Preparing activity:
Navy-TD
Project No.

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MIL-G-29011(TD)
1 March 1971
SUPERSEDING
(See 6.2)

MILITARY SPECIFICATION

GUIDES, OPERATION AND MAINTENANCE, FOR TRAINING AIDS, FOR PREPARATION OF

This specification has been approved by the Naval Training Device Center, Department of the Navy.

1. SCOPE

1.1 This specification covers the requirements for the preparation of operation and maintenance guides for classroom training aids such as cutaways, take-aparts, animated display panels, panel boards, animated transparencies, and other instructional aids of similar nature.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-M-38784	Manual, Technical; General Requirements for Preparation of
MIL-M-81203	Manual, Technical; In-Process Reviews, Validation, and Verification Support of

PUBLICATIONS

Department of Defense

DLSC H4-1	Federal Supply Code for Manufacturers

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

TMSS
AMSC N3483

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3. REQUIREMENTS

3.1 General manner of preparation.- The general preparation requirements shall be in accordance with MIL-M-38784.

3.1.1 General.- When directed by the procuring activity, guides for animated transparencies shall be prepared in accordance with Appendix A. When directed by the procuring activity, guides for training aids which require a minimal amount of operation and maintenance information shall be prepared in accordance with Appendix B.

3.1.2 Security classification markings.- Security classification markings for the operation and maintenance guide shall be in accordance with MIL-M-38784.

3.2 Content and arrangement.- The operation and maintenance guide shall have the following sections in the sequence shown:

(a) Front Matter (See 3.2.1)

(b) Section I, General Description (See 3.2.2)

(c) Section II, Installation, Adjustment and Operation (See 3.2.3)

(d) Section III, Maintenance (See 3.2.4)

(e) Section IV, Supplementary Information (See 3.2.5).

3.2.1 Front matter.- The front portion of the guide shall consist of the following items prepared in accordance with MIL-M-38784:

(a) Cover/Title Page

(b) List of effective pages

(c) Table of contents

(d) List of illustrations

(e) List of tables

(f) Safety Summary.

3.2.2 Section I, general description.- Section I of the guide shall provide the requirements specified in 3.2.2.1 and 3.2.2.2 of this specification.

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3.2.2.1 Frontispiece.- A frontispiece for the guide (usually a left-hand page facing the first page of Section I and identified as figure 1-1) shall be prepared as a composite illustration of all equipment which comprises the training aid, giving proper nomenclature of the components indexed and captioned in accordance with the requirements of MIL-M-38784. Other equipment used with, but not supplied as part of, the training aid, including operational equipment, shall be shown and indexed but shall be marked "not supplied".

3.2.2.2 Description.- A brief, comprehensive review of the training aid, in nontechnical, easily understood terms, shall be provided, including, but not limited by, the following:

(a) The overall identification of the training aid by Government and manufacturer's type designations, with differences in models if applicable

(b) The purpose of the training aid

(c) The extent to which the training aid illustrates or demonstrates the operation or principle for which it was designed

(d) The level of training which the training aid was designed to support

(e) A physical description, including the overall size (width, length, and height) and the weight of the training aid

(f) Power requirements, including any special instructions regarding power, cables, linkages, or other details

(g) A description of the functional operation of the training aid components

(h) A list of equipment (including accessories such as cables, linkages, maintenance tools and test equipment, and publications) supplied under the training aid contract

(i) A list of equipment (including operational equipment, and accessories such as cables, linkages, maintenance tools and test equipment) required but not supplied under the training aid contract.

3.2.3 Section II, installation, adjustment and operation.- Section II of the guide shall provide the requirements specified in 3.2.3.1 through 3.2.3.3 of this specification.

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3.2.3.1 Installation Instructions.- Installation instructions shall be complete, as required by the complexity of the training aid, and shall include:

(a) Location of desirable positions for the installation of the training aid

(b) Precautionary measures and checks necessary during installation, adjustment, and operation to prevent faulty operation or damage

(c) Suggested means for handling and storing the training aid.

3.2.3.2 Adjustment.- The necessary adjustments shall be listed in sequential order and shall be described.

3.2.3.3 Operating instructions.- Complete instructions shall be given on how to operate or handle the training aid in order to accomplish each of its functions.

3.2.4 Section III, Maintenance.- Section III of the guide shall contain all information necessary for maintenance of the training aid, including instructions for inspection, preventive maintenance, and repair. Except as stated in 3.2.5.2, all drawings pertaining to maintenance shall be located in this section. Text of this section shall include, as required by the complexity of the training aid, but not be limited by, the following:

(a) Inspection procedures

(b) Cleaning instructions, including cleaning agents to be used and precautions to be observed

(c) Lubricating instructions, including the kind of lubricants required, methods of applications, and frequency

(d) Instructions to permit fault isolation of electrical and electronics components. Appropriate references shall be made to diagrams located in Section IV

(e) Complete disassembly and reassembly procedures

(f) If applicable, a statement to the effect that spare parts are not provided and must be fabricated locally

(g) A list of the types of materials (such as wood, plastic, type of plastic cement, type of paint or lacquer, etc.) to be used in repairing the training aid.

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3.2.5 Section IV, supplementary information.- Section IV of the guide shall provide the requirements specified in 3.2.5.1 through 3.2.5.3 of this specification.

3.2.5.1 List of manufacturers.- The manufacturers' Federal Supply Codes, names and addresses shall be listed numerically by code number as listed in DLSC H4-1, as shown in figure 1 of this specification.

3.2.5.2 List of maintenance parts.- Maintenance parts shall be listed, including all electrical and electronic components and all mechanical components that are subject to wear or deterioration. Noun name, modifier, basic characteristics, part number, manufacturer's code, and reference designator or figure and index number of the part shall be included when applicable. Electrical and electronic components shall be listed first in alphanumeric order by reference designator, followed by all other parts listed in alphanumeric order by part number. The list shall be prepared as shown in figure 2 of this specification.

3.2.5.3 Diagrams.- When the training aid is electrical, wiring diagrams shall be provided to show the origin of, all tie points for, and termination of, each lead. In addition, applicable schematic diagrams shall be included. Diagram preparation shall be in accordance with MIL-M-38784.

4. QUALITY ASSURANCE PROVISION

4.1 The quality assurance provision shall be in accordance with MIL-M-38784 and as specified in 4.1.1 of this specification.

4.1.1 Validation and verification shall be in accordance with MIL-M-81203.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-M-38784.

6. NOTES

6.1 Intended use.- This specification covers the requirements for the preparation of guides for the installation, operation and maintenance of training aids.

6.2 Supersession data.- This specification supersedes the following specifications:

NAVAL TRAINING DEVICE CENTER (NAVTRADEVCEN)

422-111	Animated Transparency Guides and Summaries; Specification for, dated 1 September 1965

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Order of Precedence. In the event of a conflict between the

422-120	Preparation of Guides for Classroom Training Aids, such as Cutaways, Take-Aparts, Panel Boards; General Specification for, dated 16 March 1964
422-121	Instructor's Guides for Animated Panel Training Aids; General Specification for, dated 16 March 1964

CUSTODIAN:
Navy-TD

PREPARING ACTIVITY:
Navy-TD
Project No. TMSS-N043

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LIST OF MANUFACTURERS

CODE	MANUFACTURER	ADDRESS
01121	Allen-Bradley Company	1201 South 2d Street Milwaukee, Wisconsin 53204
02660	Amphenol Corporation	2801 South 25th Avenue Broadview, Illinois 60153
02799	Arco Capacitors, Incorporated	1548 South Robertson Los Angeles, Calif. 90035
03508	General Electric Company Semi-Conductor Products Dept.	Electronics Park Syracuse, New York 13201
08806	General Electric Company Miniature Lamp Department	Nela Park Cleveland, Ohio 44112
16017	KGS Electronics	157 Winston Way Pasadena, Calif. 91107
35434	Lectrohm Incorporated	5560 Northwest Highway Chicago, Illinois 60630
63743	Ward Leonard Electric Co.	30 South Street Mount Vernon, New York 10550
71823	Darnell Corporation, Ltd.	Downey, Calif. 90241
91929	Honeywell, Inc. Microswitch Division	Chicago and Spring Streets Freeport, Illinois 61032

Figure 1. List of Manufacturers (Sample)

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LIST OF MAINTENANCE PARTS

REF. DESIG. or FIG. and INDEX NO.	PART NAME AND DESCRIPTION	PART NUMBER	MFR CODE
CR1	Semiconductor Device, Diode	1N91	
CR10	Semiconductor Device, Diode	1N538	
CR11	Semiconductor Device, Diode	1N91	
CR2	Semiconductor Device, Diode	1N433	
CR3	Semiconductor Device, Diode	1N91	
CR4	Semiconductor Device, Diode	1N538	
CR5 thru CR9	NOT USED		
C1	Capacitor, Fixed, Paper, .047 UF, 400V, 20 PCT	CPO5A1EE473M	
C2	Capacitor, Fixed, Plastic, .01 UF, 400VDC	L34-103	99120
C3	Capacitor, Fixed, Electrolytic, 10 UF, 100V	NLW10-100	14655
DS1	Lamp, .04W, 125V, Glow, Neon Gas	NE51	24446
J1	Connector, Receptacle, Electrical, 37 Pins	MS3100R28-21P	96906
J2	Jack, Telephone, Bushing, Mtd	S-11	82389
K1	Relay, Time Delay, Adjustable, 5 PDT, 15 AMPS, 115V, 60 Hz	ND12	70382
Q1	Transistor, NPN Grown Junction	2N1152	96124
R1	Resistor, Fixed, Comp, 220 Ohms, 5 PCT, 1/2W	MS35043-71	
R10	Resistor, Fixed, Comp, 100K, 5 PCT, 1/2W	MS35043-135	
R2	Resistor, Variable, 5K, 1 PCT, Linearity 0.3 PCT	200CEUS1109	98009
R3	Resistor, Variable, 500 Ohms, 10 PCT, 2W	JJS	01121
R4	Resistor, Fixed, Comp, 10K, 5 PCT, 1W	RC32GF103J	
R5 thru R9	Resistor, Fixed, Comp, 1 Meg, 5 PCT, 1/2W	MS35043-159	
S1	Switch, Sensitive, 15 AMP	BZ-2RS	91929
3-2(4)	Level, Bubble	230BE2	39482
1-4(3)	Filter, Air Line	3037	45681

Figure 2. List of Maintenance Parts (Sample)

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APPENDIX A

GUIDES, OPERATION AND MAINTENANCE, FOR ANIMATED TRANSPARENCIES

10. SCOPE

10.1 This appendix contains the content, style and format requirements for the preparation of operation and maintenance guides for animated transparency training aids.

10.2 Security classification markings.- Security classification markings for the operation and maintenance guide shall be in accordance with MIL-M-38784.

20. APPLICABLE DOCUMENTS

20.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

PUBLICATIONS

United States Government Printing Office

Library of Congress Catalog No. Z 253, U58 — GPO Style Manual

(The above publication is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

30. REQUIREMENTS

30.1 Text content.- The guide for animated transparency shall be a one-or two-page (single sheet) publication containing all necessary information for using the subject animated transparency. (See figure A-1.) Unless otherwise specified in writing by the procuring activity, no photograph or other artwork shall be used in the guide. The following main topics and such subtopics as may be required by the nature of the animated transparency shall be included. These topics shall be numbered as specified in 30.2.1.1 and shall be presented in the sequence shown.

(a) General (See 30.1.1)

(b) Description (See 30.1.2)

(c) Application (See 30.1.3)

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(d) Maintenance (See 30.1.4)

(e) Supplementary Information (See 30.1.5)

(f) Government's rights in data (See 30.1.6).

30.1.1 General.- This topic shall be developed as specified in 30.1.1.1 and 30.1.1.2 of this specification.

30.1.1.1 Standard statements.- Subparagraphs a, b, and c, and the caution under the heading "General" shall be standard statements. Except for the quotation marks, these items shall appear in the exact words, as follows, arranged and underlined as shown:

"(a) Use this training aid with an overhead projector such as Device 1BC7, Master Vu-Graph, C. Beseler Company, No. 6600, or equivalent, using a 1,000 watt bulb.

"(b) To remove animated transparency: Place carrying case on table with the side marked "TOP" up; open case and carefully lift transparency by grasping "LIFT" knobs or bars which are permanent parts of the device

"(c) To project transparency: Place transparency on projector stage so that alinement pins are engaged in the two holes provided in the projector.

"CAUTION: If wattage of projection lamp exceeds the rated wattage of the projector in use, the animated transparency is likely to be damaged by heat."

30.1.1.2 Standard heading.- Subparagraph d under the heading "General" shall include the following standard heading; except for the quotation marks, the heading shall appear in the exact words as follows, underlined as shown:

"(d) To replace animated transparency in its carrying case:"

30.1.1.3 Subparagraph d shall list proper positioning of all controls and special instructions covering replacement of the animated transparency in its case. This item shall be developed under subparagraphs numbered (1), (2), etc., as shown in figure A-1 (Sheet 1 of 2).

30.1.2 Description.- This topic shall provide a brief, comprehensive description of the animated transparency. This description shall provide, wherever applicable, the requirements specified in 30.1.2.1 through 30.1.2.4, but shall not be limited thereto.

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30.1.2.1 Identification shall be made of the equipment, component, assembly, or principle that the animated transparency is designed to illustrate, demonstrate or otherwise present for training purposes.

30.1.2.2 Mention shall be made as to how the animated transparency is to be used (e.g., to illustrate a lecture, to show relationships of components, or other use).

30.1.2.3 A statement shall be made regarding the extent to which the animated transparency depicts the operation or principles for which it was designed to present.

301.2.4 Controls.- Each control shall be listed with a brief description thereof. This description shall include the control number or letter as engraved on the animated transparency; the direction of movement (left or right, towards top or front, clockwise or counterclockwise); and the related functional action. Direction of movement shall be given to prevent damage to the transparency and shall conform to the reference points of LEFT, RIGHT, TOP and FRONT as engraved on the baseplate of the transparency.

30.1.2.5 Color coding.- Each projectable component or component assembly, shall be listed opposite the name of the color used therefor.

30.1.3 Application.- This topic shall be developed to include the following:

(a) Any cautions required to prevent damage to the animated transparency

(b) Brief explanation of the basic demonstration(s) made possible by the use of the animated transparency

(c) Any preliminary instructions necessary for positioning of components prior to each demonstration

(d) Step-by-step instructions for using the animated transparency in a demonstration, including brief explanation of what each major action or change in the projection represents or portrays

(e) Any specific steps, timing, or particular aspects of procedures required by features peculiar to the animated transparency.

30.1.3.1 Instructions shall be definite and direction of movement shall be indicated by the reference points of LEFT, RIGHT, TOP, and FRONT as engraved on the baseplate of the animated transparency. Verbs shall be specific, appropriate to the type of control, and consistent throughout the guide.

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30.1.4 Maintenance.- This topic shall provide maintenance instructions for the animated transparency. The following statements, omitting quotation marks, shall be included in the exact words, arranged and underlined as shown:

"(a) Inspection: Perform routine visual inspection for loose or broken parts prior to projection demonstration.

"(b) Cleaning: With a dry, lintless cloth, carefully wipe dust and finger marks from the plastic parts of the transparency. Do not use liquids or abrasive materials for any purpose.

"(c) Replacement parts are not supplied for the animated transparency. Fabricate any necessary part from clear, acrylic plastic of the correct color, using the original part as a template. If damage is beyond local repair capabilities, contact the Naval Training Device Center (Code 431), Orlando, Florida 32813 for information."

30.1.5 Supplementary information.- This topic shall list text reference material, as well as any related training aids or other equipment not listed elsewhere in the guide, that would be helpful to the instructor.

30.1.6 Government's rights in data.- The following copy shall be boxed within a 1-point line, shall be unnumbered, and appear as the last item of text:

"Reproduction of this publication in whole or in part is permitted for any purpose of the United States Government."

30.2 Style and format.- Style and format requirements for the preparation of manuscript copy and reproducible text of the guide shall be as specified in 30.2.1 through 30.2.4 of this specification.

30.2.1 Text preparation.- Text preparation shall be as specified in 30.2.1.1 through 30.2.1.9.

30.2.1.1 Numbering and indenting paragraphs.- Main topics shall be numbered with Arabic numerals followed by a period. Subparagraphs thereunder shall be identified by lower-case letters of the alphabet followed by a period. Subordinate breakdowns under these paragraphs shall be numbered with Arabic numerals with parentheses. Any further subordination shall be identified by lower-case letters of the alphabet within parentheses. Each subordination shall be indented as shown in figure A-1.

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30.2.1.2 Level of writing.- Text shall be written for understanding by military enlisted technicians who have completed high school. Technical phraseology shall be avoided.

30.2.1.3 Grammatical treatment.- The second person imperative shall be used for operational procedures. For example, "Move Control No. 3 to the extreme left." The third person shall be used for description and discussion. For example, "The sear is disengaged from the collar of the percussion mechanism." (See 30.1.3.1).

30.2.1.4 Nomenclature consistency.- Nomenclature shall be consistent throughout each publication and throughout directly related publications. Terms used in the guide shall be consistent with the terms used therefor in the technical manual(s) for the operational equipment portrayed by the animated transparency.

30.2.1.5 GPO style manual.- The U.S. Government Printing Office Style Manual shall be used as a general guide for spelling, compound word forms, capitalization, punctuation, and abbreviations.

30.2.1.6 Numerical expressions.- Figures shall be used for isolated numbers of 10 or higher. Numbers less than 10 shall be spelled out except when used in a group of two or more related numerical expressions, or when used as a unit of quantity or measurement. Double statement of numbers (number spelled out followed by figure in parentheses) shall not be used. Except when describing the caliber of ordnance weapons, numbers less than unity involving decimals shall begin with a zero before the decimal point.

30.2.1.7 Device numbers.- Device numbers shall not be hyphenated, and the alphabetical elements thereof shall be capitalized, e.g., Device 2H125.

30.2.1.8 Abbreviations.- Abbreviations shall be held to a minimum. Unless otherwise approved in writing by the procuring activity, only those abbreviations listed in Chapter 10 of GPO Style Manual shall be used.

30.2.1.9 Cautions and notes.- Use of these adjuncts to text shall be as sparing as is consistent with real need. (Cautions are used to emphasize or caution against operating procedures, practices, and such, in order to prevent damage to equipment. Notes are used to highlight an operating procedure, condition, or such.) When used, cautions shall immediately precede the applicable instructions.

30.2.2 Manuscript copy.- Manuscript copy shall be typed double spaced on one side of the sheet only, and be furnished in the quantity indicated by the procuring activity. The manuscript shall be suitable for review for technical accuracy, scope and content, style, grammar, readability, method of treatment, general layout, and compliance with this specification.

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30.2.3 Reproducible copy.- The reproducible copy shall reflect all corrective action resulting from review of the manuscript copy. The reproducible copy shall be typed on one side only of oversize white bond paper, and shall be suitable for offset reproduction. Typing errors shall be neatly corrected by mortising in the correction using white tape on the back of the reproducible copy. Typing shall be single spaced, double spaced between paragraphs if space permits, and in two columns of approximately equal length. (See figure A-1.) Right-hand margins shall be unjustified. Type shall be IBM Boldface No. 1 or an equivalent proportional-spaced type face that will scale approximately 7 lines to the inch when reduced 20 percent.

30.2.3.1 Dimensions.- (See figure A-2.) Unless approval to vary dimensions is obtained in writing from the procuring activity prior to preparation of the reproducible copy, the following dimensions shall apply:

(a) Overall image area: 12-5/8 inches horizontal by 9-5/16 inches vertical

(b) Column width: 6-1/8 inches

(c) Column length: approximately 8 inches (full length column after allowing for title block and 3/8-inch space between last line of block and first line of text)

(d) Space between columns: 3/8 inch.

30.2.3.2 Title block of guide.- Unless otherwise specified by the procuring activity, the title block of the guide shall be as shown in figure A-2. The publication number will be furnished by the procuring activity at time of approval of the manuscript copy.

30.2.3.3 Reverse side of guide.- (See figure A-1 (Sheet 2 of 2).) The second page of the reproducible copy shall be prepared in the same dimensions as the first page. The title block shall be omitted from the top, and text shall continue in two approximately equal columns.

30.2.3.3.1 If space permits, the title of the guide, the device number, publication number, and name and address of the procuring activity shall appear at the bottom center of the reverse side of the guide. If space permits, the procuring activity's slogan, provided at time of approval of the manuscript copy, shall appear centered under these items.

30.2.4 Revised guides.- When a revised guide is a deliverable item of contract, the requirements specified herein shall apply except that the revised guide shall be identified by the word "Revised" (in upper and lower case) followed by the date of the revision (month and year), to appear in place of the original date, e.g., Revised February 1971.

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OPERATION AND MAINTENANCE GUIDE FOR
ANIMATED TRANSPARENCY, SOLION UNIT, PRESSURE DETECTOR MK 1 MOD O (MK 52/55 MINES)

DEVICE 3C171 — NAVTRADEV P-9532

NAVAL TRAINING DEVICE CENTER
ORLANDO, FLORIDA 32813

February 1971

UR-NAME COMPANY, INC.
CONTRACT N61339-71-C-9822

1. GENERAL

a. Use this training aid with an overhead projector such as Device 1BC7, Master Vu-Graph, C. Beseler Company No. 6600, or equivalent, using a 1000 watt bulb.

b. To remove animated transparency: Place carrying case on table with the side marked "TOP" up; open case and carefully lift transparency by grasping "LIFT" knobs which are permanent parts of the device.

c. To project transparency: Place transparency on projector stage so that alinement pins are engaged in the two holes provided in the projector.

CAUTION: If wattage of projection lamp exceeds the rated wattage of the projector in use, the animated transparency is likely to be damaged by heat.

d. To replace animated transparency in its carrying case:

(1) Check that Control No. 1 (flap) is down.
(2) Check that Control No. 2 (flap) is down.
(3) Check that Control No. 4 (flap) is down.
(4) Check that Control No. 6 (flap) is down.

2. DESCRIPTION

a. General: This device is an animated transparency which demonstrates the principle of operation and the function of the Solion Unit as a part of the Pressure Detector MK 1 MOD O (MK 52/55 Mines).

b. Action Demonstrated: The device illustrates the Solion's function of converting external hydraulic pressure into electrical energy by means of a chemical process, and in transmitting a signal.

c. Controls: Six controls are used to operate the device. They are identified as follows:

Control	Description and Function
No. 1	Lifted or lowered to expose or conceal negative pressure arrow
No. 2	Lifted or lowered to expose or conceal pressure arrow
No. 3	Rotated counterclockwise; controls ion movement
No. 4	Lifted or lowered to expose or remove current arrows
No. 5	Moved toward "TOP" or "FRONT" to control indicator meter
No. 6	Lifted or lowered to depict or remove direction of current flow

d. Color Coding: To facilitate identification during demonstrations, projected parts are colored as follows:

Pink	Positive electrode, circuit and meter
Red	Current flow arrows
Dark Grey	Negative pressure arrow, negative circuit lines
Medium Grey	Negative ions
Light Grey	Iodine solution
Dark Green	Center wall
Light Green	Diaphragms
Dark Blue	Housing wall
Light Blue	Pressure and arrow
Black	Negative electrode, positive ions

3. APPLICATION

a. Basic Demonstration: The following demonstration shows the operations of the Solion Unit as part of the Pressure Detector MK 1 MOD O (MK 52/55 Mines).

Figure A-1. Guide for an Animated Transparency (Sample) (Sheet 1 of 2)

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b. Preliminary Proce — Before each demonstration:

(1) Check that Cont — o. 1 (flap) is down
(2) Check that Cont — o. 2 (flap) is down
(3) Check that Cont — o. 3 is positioned so start
arrows line up
(4) Check that Cont — o. 4 (flap) is down
(5) Check that Cont — o. 5 is at extreme "FRONT"
(6) Check that Cont — o. 6 (flap) is up

c. Demonstration Pro — e:

(1) Raise Control N — (flap), exposing negative pressure
arrow, then raise — ol No. 2 (flap) exposing arrow and
word "PRESSURE" — ating the pressure created by the
negative pressure.

(2) Rotate Control — counterclockwise until moving
ions contact negati — ctrode. At this time the ions
will change from n — n grey to black, indicating a
chemical change w — y the ion picks up an electron and
becomes negatively — ged.

(3) Turn Control N — again until the ion contacts the
positive electrons, — ich time it will turn back to
medium grey. Thi — ther exchange of electrons takes
place completing th — ctric circuit.

(4) Lift Control No — lap) to expose the circuit arrows
indicating an electr — w through the circuit. Now push Control
No. 5 towards "TC — til it stops, thus moving the
indicator needle on — neter to register the presence of
an electric current

(5) Continue to tur — trol No. 3 until the ion contacts
the next negative e — de. The ion will again turn black,
indicating an electr — change (chemically).

(6) Turn Control N — again until the ion contacts the
next positive electr — here it turns back to medium
grey showing a dis — e of electrons and completing an
electric circuit.

(7) Lower Control No. 6 (flap) to expose direction of current flow of electricity.

(8) At end of demonstration, lower Controls No. 1, 2 and 4; push Control No. 5 down to extreme front. The animated transparency is now ready for storage.

4. MAINTENANCE

a. Inspection: Perform routine visual inspection for loose or broken parts prior to projection demonstration.

b. Cleaning: With a dry, lintless cloth, carefully wipe dust and finger marks from the plastic parts of the transparency. Do not use liquids or abrasive materials for any purpose.

c. Replacement parts are not supplied for the animated transparency. Fabricate any necessary part from clear acrylic plastic of the correct color, using the original part as a template. If damage is beyond local repair capabilities, contact the Naval Training Device Center (Code 431), Orlando, Florida 32813 for information.

5. SUPPLEMENTARY INFORMATION

a. Reference Publications:

NAVWEPS OP 2567, Firing Mechanisms (Confidential)

b. Related Animated Transparencies:

(1) Display Panel Control Box (MK 52/55 Mines), Device 3C170.

(2) Transparencies Set and Charts (MK 52/55 Mines), Device 3C172.

(3) Animated Display Panel (MK 52/55 Mines), Device 3C174.

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GUI — DEV — R ANIMATED TRANSPARENCY, SOLION UNIT, PRESSURE DETECTOR MK 1 MOD O (MK 52/55 MINES)

NAVAL TRAINING DEVICE CENTER
ORLANDO, FLORIDA

Figure 1. Guide for an Animated Transparency (Sample) (Sheet 2 of 2)

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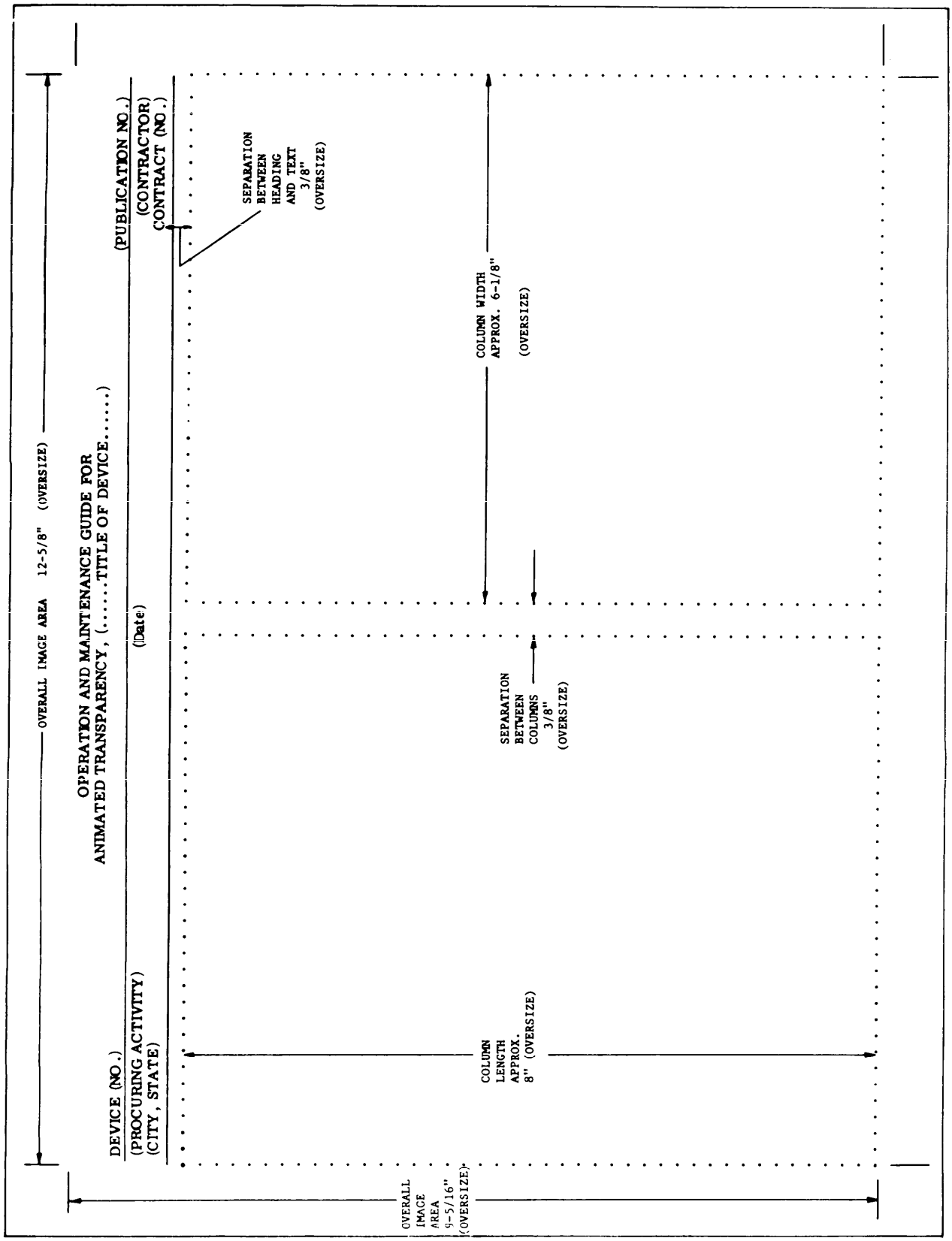


Figure A-2. Oversize Layout Sheet for Guide, Scaled for 20% Reduction (Sample)

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APPENDIX B

GUIDES, OPERATION AND MAINTENANCE,
FOR SIMPLE TRAINING AIDS

10. SCOPE

10.1 This appendix covers the content, style and format requirements for the preparation of operation and maintenance guides for simple training aids which require a minimal amount of operation and maintenance information.

10.2 Security classification markings.- Security classification markings for the operation and maintenance guide shall be in accordance with MIL-M-38784.

20. APPLICABLE DOCUMENTS

20.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

PUBLICATIONS

United States Government Printing Office

Library of Congress Catalog No. Z 253, U58 — GPO Style Manual

(The above publication is available from the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.)

30. REQUIREMENTS

publication containing all necessary information for using the training aid. (See figure B-1.) The main topics specified in 30.1.1 through 30.1.16 of this specification and such subtopics as may be required by the nature of the training aid shall be included. An overall photo or other illustration of the device or two illustrations if necessary to convey the general idea shall be included. The size and shape of illustrations may vary as required.

30.1.1 Training category.- This shall be as furnished by the procuring activity at time of approval of the manuscript copy.

30.1.2 Originating agency.- This shall be as furnished by the procuring activity at time of approval of the manuscript copy.

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30.1.3 Security classification of device.- This shall be a statement giving the security classification of the device, amplified as necessary, for example, where accessories may be classified and the device itself unclassified.

30.1.4 Purpose of device.- In one or two sentences, the training purpose of the device shall be summarized, so that the reader can determine at a glance if the device has application to a training problem at his activity. In effect, this statement shall answer the question, "what training problem does the device help to solve?"

30.1.5 Intended use.- This paragraph shall state where the device is to be used (e.g., in flight, on shipboard, in classroom, on target range); type of school; for on-the-job or in-service training (in what type of activity(s); the level of training (basic or advanced); the type of trainee (e.g., pilot, maintenance man, math student, navigator, gunner, radar operator, sonar operator, etc.); rank or rate of trainee or any special prerequisites which trainee should have accomplished prior to use of device.

30.1.6 Functional description.- The functional description shall briefly state the type of device, how it is used, inherent training features, and how it achieves its functional purpose (how it works). The functional description shall answer in as few words as possible the applicable questions listed below:

(a) What is the device? What does it look like? Of what does it consist (name of the panels, mockups, etc.)?

(b) How is the device used? Does it illustrate, visualize, or demonstrate? What are the instructional uses? For what specific area of training? To teach what information or develop what skills? For specifically what operational equipment? What techniques?

(c) What are the inherent training features? Can the speed of presentation be controlled? Can the sequence of events be varied?

(1) What is there about the device which in itself will attract interest of the trainee? Is the training objective of the device obvious from its appearance and if so, how?

(2) Can the device be used to stimulate competition of trainees with their own records, with other trainees, or between two or more groups in the class and if so, what features do this?

(3) If the device is trainee operated, does it lend itself to repetition and if so, what can be repeated?

(d) How does the device achieve its functional purposes (i. e., how does it work)?

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30.1.7 Physical information.- This topic shall state size and weight of the device. It shall list any accessories furnished, such as charts, tapes, records, films.

30.1.8 Equipment required (not supplied).- Any equipment needed or desired for optimum utilization of the device shall be listed, stating whether it is required or desired. If none is needed or desired, the word "None" shall appear under this subhead.

30.1.9 Power requirements.- External power requirements for the device shall be stated. If no power is required, the word "None" shall appear under this subhead.

30.1.10 Reference publications not supplied.- This information shall be as furnished by the procuring activity. If not applicable, this subhead shall be omitted entirely.

30.1.11 Personnel.- Personnel data for the device, as applicable, shall be itemized under this heading. E.g.,

Instructor(s):	Number, type, and qualifications
Trainees:	Number accommodated by device

30.1.12 Operation.- A paragraph(s) giving any installation or operating instructions required, including any cautions necessary to avoid damage to equipment or warnings necessary to prevent injury to personnel, shall appear under the subhead "Operation".

30.1.13 Maintenance.- A paragraph(s) giving any necessary maintenance information, including any cautions or warnings required, as well as any required information on obtaining or making replacement parts shall appear under the subhead "Maintenance".

30.1.14 Related training devices.- This information shall be as furnished by the procuring activity. If not applicable, this subhead shall be omitted entirely.

30.1.15 Contract identification.- The name and address of the manufacturer and the complete contract number under which the device was manufactured shall be stated. (See figure B-1 (Sheet 2 of 2).) The following copy shall be boxed within a 1-point line and appear as the last item of text:

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30.2 Style and format.- Except for the changed requirements of 30.2.1 through 30.2.2.3, the style and format requirements of Appendix A shall apply.

30.2.1 Numbering and indenting paragraphs.- Main topics shall be unnumbered. Paragraphs thereunder shall be indented as shown in figure B-1.

30.2.2 Reproducible copy.- The reproducible copy shall reflect all corrective action resulting from review of the manuscript copy. The reproducible copy shall be typed on one side only of oversize white bond paper, and shall be suitable for offset reproduction. Typing errors shall be neatly corrected by mortising in the correction using white tape on the back of the reproducible copy. Typing shall be double-column, single spaced, and otherwise as shown in figure B-1. Right-hand margins shall be unjustified. Type shall be IBM Boldface No. 1 or an equivalent proportional-spaced type face that will scale approximately 7 lines to the inch when reduced 20 percent. Position of the halftone(s) shall be indicated on the layout by a box outlined in blue pencil. Line artwork shall be reduced to the appropriate size and pasted directly onto the reproducible layout and shall be boxed within a 1-point line.

30.2.2.1 Dimensions.- The oversize reproducible image shall not exceed 7-3/4 by 11-3/4 inches, typed on paper approximately 10 by 14-1/2 inches.

30.2.2.2 Title block of guide.- Unless otherwise specified by the procuring activity, the title block of the guide shall be as shown in figure B-1 (Sheet 1 of 2). The publication number will be furnished by the procuring activity at time of approval of the manuscript copy.

30.2.2.3 Artwork.- Artwork shall be prepared in accordance with the requirements of MIL-M-38784.

30.2.2.3 Deliverable items.- Deliverable items of reproducible copy shall consist of:

(a) Reproducible layout sheets (which include text and any line artwork involved)

(b) Glossy photo intended for halftone illustration (mounted, cropped and sized).

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OPERATION AND MAINTENANCE GUIDE FOR
TRAINING AID, MK V IRVIN BAROMETRIC RELEASE MODEL

DEVICE 9F90

NAVTRADEV P-8723
February 1971

NAVAL TRAINING DEVICE CENTER — ORLANDO, FLA.

TRAINING CATEGORY:

SURVIVAL TRAINING (Parachute)

ORIGINATING AGENCY: DCNO (Air)

SECURITY CLASSIFICATION OF DEVICE:

Device 9F90 is unclassified.

PURPOSE OF DEVICE:

a. To identify parts, show relationship between parts, and demonstrate the sequence of actions in the MK V Irvin Barometric Release.

b. To provide realistic training and drill in the principles of operation, adjustment, and maintenance of the MK V Irvin Barometric Release.

INTENDED USE:

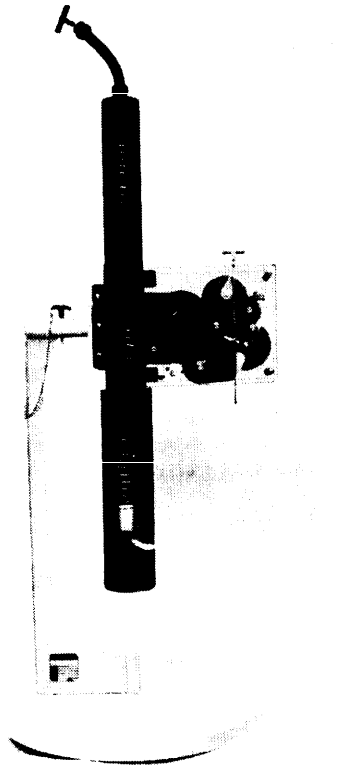
In Parachute Rigger's Schools or Aviation Physiological Training Units for classroom training of enlisted Airmen or Parachute Riggers with little or no background in automatic release devices.

FUNCTIONAL DESCRIPTION:

Device 9F90 is a 4:1 scale, 3-dimensional, working model of the MK V Irvin Barometric Release. It is made of metal, wood, and acrylic plastic. Housings are removable, so as to reveal the gear train, compression spring, power cable, and other mechanisms. Coverplates are of clear acrylic so that the internal operations may be viewed without disassembling.

The model is mounted on a wood and metal base and may be rotated through 360 degrees horizontally or vertically, to permit easy viewing from any angle.

A helper spring, not found in the operational equipment, is used on the largest gear in the gear train. The spring, which is covered to make it as inconspicuous as possible, is used to keep the force of the main spring low enough to permit hand recocking. Each gear is inscribed with a white line to accentuate small movements.



Parts are color-coded as follows:

Yellow	Piston assembly and spring
Blue	Gear train and altitude setting
Red	Release latch
Orange	Arming assembly
Aluminum	Timing cam
Black	Aneroid
Green	Components not in operational equipment

The device may be used to demonstrate the following operating procedures and maintenance steps:

Figure B-1. Operational Maintenance Guide for a Simple Training Aid (Sample) (Sheet 1 of 2)

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Arming
Bailout
Timing adjustments and altitude settings
Disassembly
Change of timing

The instructor may repeat all or any part of the demonstration or may permit trainees to participate in the demonstrations.

PHYSICAL INFORMATION:

Number of pieces: Two
Size: Assembled device 27" x 58"
Weight: 34 lbs.

EQUIPMENT REQUIRED(NOT SUPPLIED):

None

POWER REQUIREMENTS:

None

REFERENCE PUBLICATIONS (NOT SUPPLIED):

Operation and Maintenance Overhaul Instructions with Parts Breakdown, Barometric Release, Irvin MK V, for Parachutes, NAVAER 13-15-504 (U).

PERSONNEL:

Instructor: Chief or lst Class PO, graduate of NATTC and qualified to teach in PR "A" School.
Trainees: Class of up to 30.

OPERATION:

Caution: Never operate the cocking lever or any other parts with a force in excess of 2 or 3 lbs. There are limit stops at both ends of the gear train travel. As soon as resistance is felt, indicating that the stop has been reached, relax the pressure or the plastic plate may crack.

For arming, proceed as follows:

a. Before resetting, move the altitude selector knob to its maximum altitude position. This will cause the aneroid stop to move out of the way, and will permit the associated gear to pass the stop.
b. Reset the pawl and gear assemblies by turning the reset handle clockwise to the stop.
c. Insert the arming rod.
d. Rotate device from vertical to the horizontal position.
e. Taking firm hold of the device, compress the power spring by pulling back on the power cable (red tee handle) until the power cable plunger engages the pawl.
f. Select release altitude by rotating the altitude pointer to the desired setting.

For bailout, proceed as follows:

a. Pull out the arming rod (orange tee handle) to simulate bailout.

Note: If the flight altitude is less than the preset altitude of the barometric release, the timing cycle will start immediately. If the flight altitude is higher than the preset altitude, proceed as follows:

b. Decrease the flight altitude with the aneroid control, to simulate free fall, by pulling the aneroid control, which simulates aneroid contraction. When the stop on the aneroid beam disengages the pin on the gear assembly, the gear assembly is free to move. Motion of the plunger is transferred thru the pawl and gear assembly to the escapement assembly. The escapement assembly applies inertia loading to the system and limits the rate at which the plunger travels. At the end of the time cycle, the pawl releases the plunger, at which point it is free to rapidly actuate the power cable with full spring force. Withdrawal of the power cable pulls the rip cord and deploys the parachute.

MAINTENANCE:

No maintenance other than routine cleaning is required. Clean plastic parts with a damp cloth only. Painted metal and wood parts may be cleaned with mild soap and water. Never use any type of solvent, such as carbontet, lacquer thinner, benzine. No provision is made for spare parts.

CONTRACT IDENTIFICATION:

Device 9F90 was manufactured by Model Model Co., Modeltown, N.Y., under NAVTRADEVCEN Contract N61339-71-C-9999.

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9F90

Figure B-1. Operational Maintenance Guide for a Simple Training Aid (Sample) (Sheet 2 of 2)

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inclusive display, can be found in MIL-M-87268 Manuals

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-G-29011 (TD)	Military Specification Guides, Operation and Maintenance for Training Aids for Preparation of.

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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